

Global Mortality Rates Change Analysis

By: Guruprasad Mukund

Agenda

- | | | |
|----|---|-----------------|
| 01 | Problem Statement & Topic Choice | Slide 3 |
| 02 | Datasets & Preparation Steps | Slide 4 |
| 03 | EDA Analysis | Slide
5-12 |
| 04 | Models Used | Slide 13-
24 |
| 05 | Main Findings & Limitations | Slide
25-26 |
| 06 | Reflection | Slide 27 |

Problem Statement

Can we investigate how mortality rates changed across countries from 2000 to 2019 based on changes in GDP per capita, health expenditure per capita, and % of the population that is age 65+?

Why this Topic?

- Interested in life insurance
- Coding project relevant to industry



Datasets & Preparation Steps

GDP Per Capita (\$) for Countries

Current Health Expenditure Per Capita (\$) for Countries

% of Population Age 65+ for Countries

Death Rate for Countries

Normalization Process

0 NF → 1 NF

- 0 NF → Wide format = each row had one country matched with 2 year values
- 1 NF → Transform to long format = every row with country matched to only 1 year's value

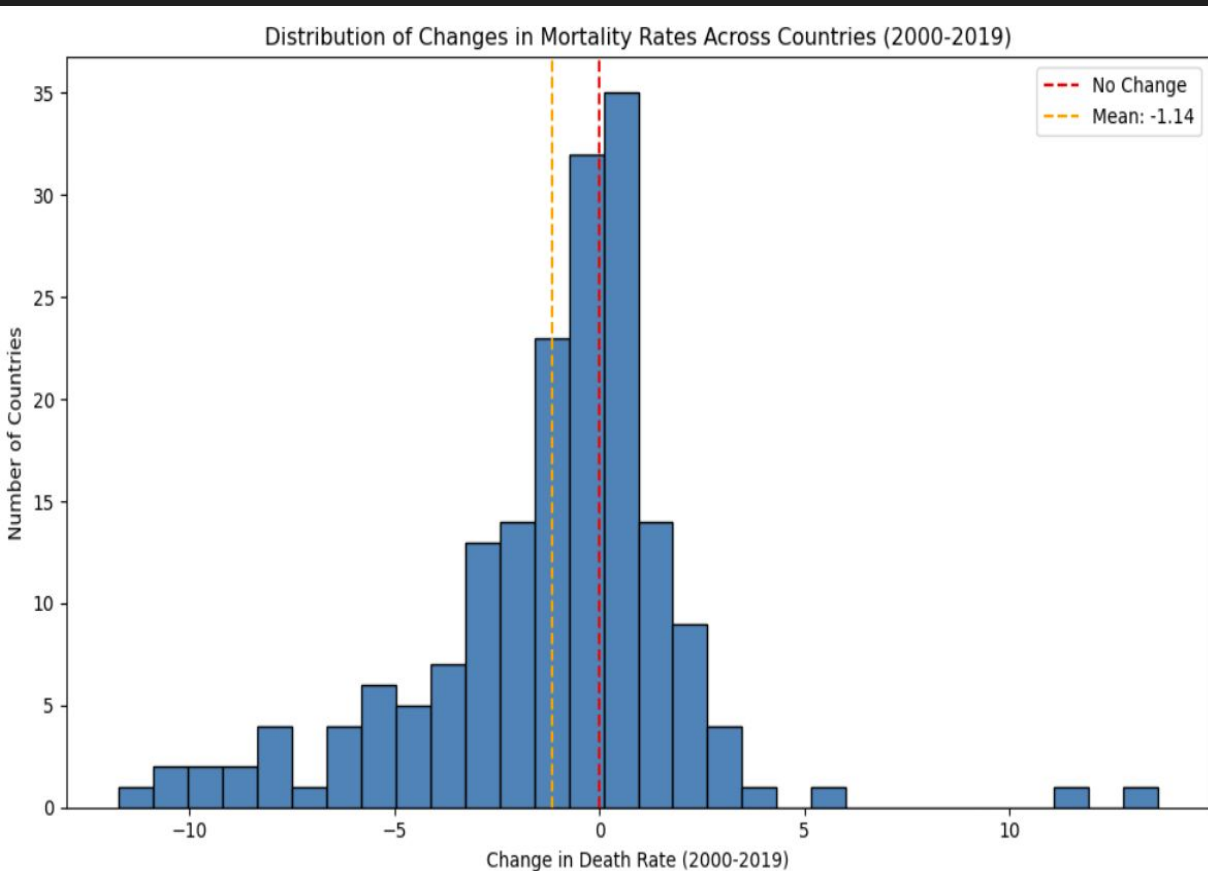
Joins, 2NF & 3NF

- Composite primary key (country_code, year)
- Partial dependency
- Predictors & dependent variable depend on composite primary key
- Country name only depends on country_code
- No 2NF, no 3NF

Staying in 1 NF

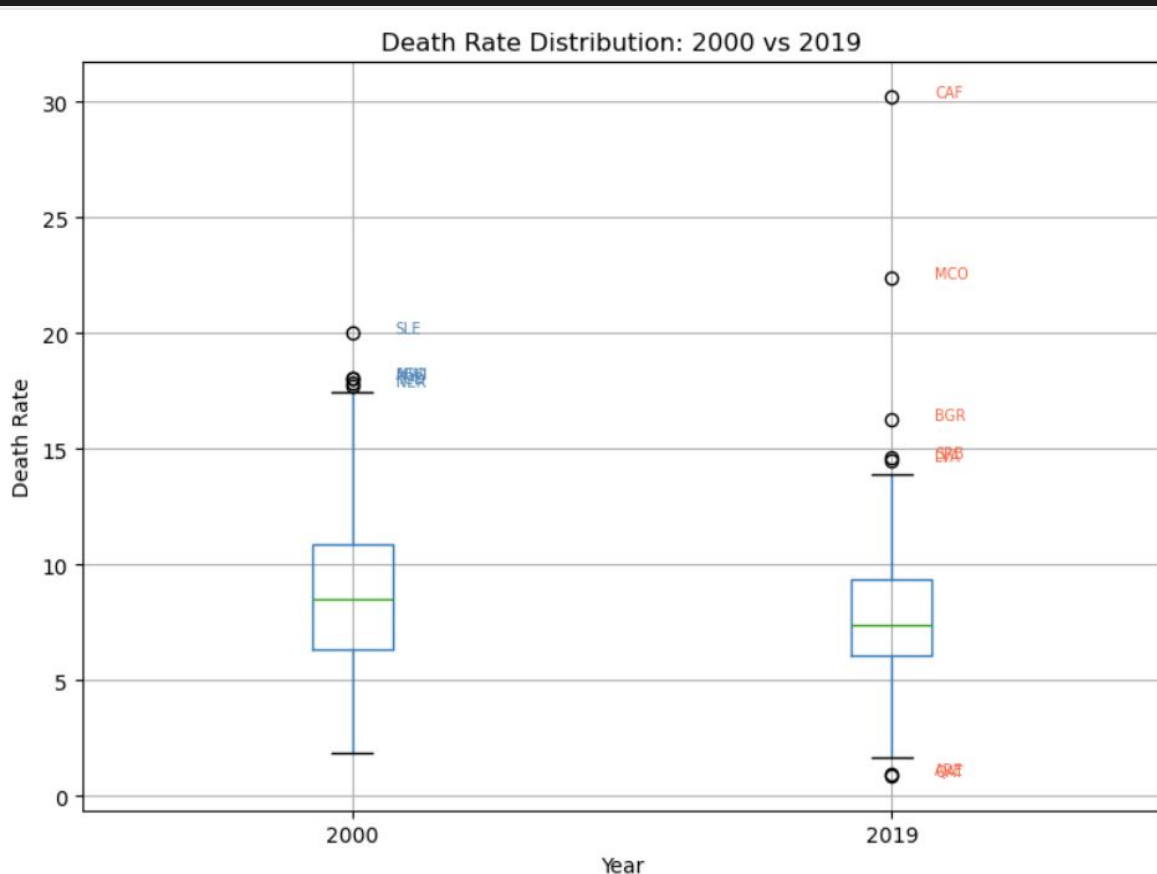
- Redundancies, but...
- Improves readability for analysis
- Includes country name
- Easier to create graphs, analyze trends

Graph 1: Histogram with Changes in Mortality Rates (2000–2019)



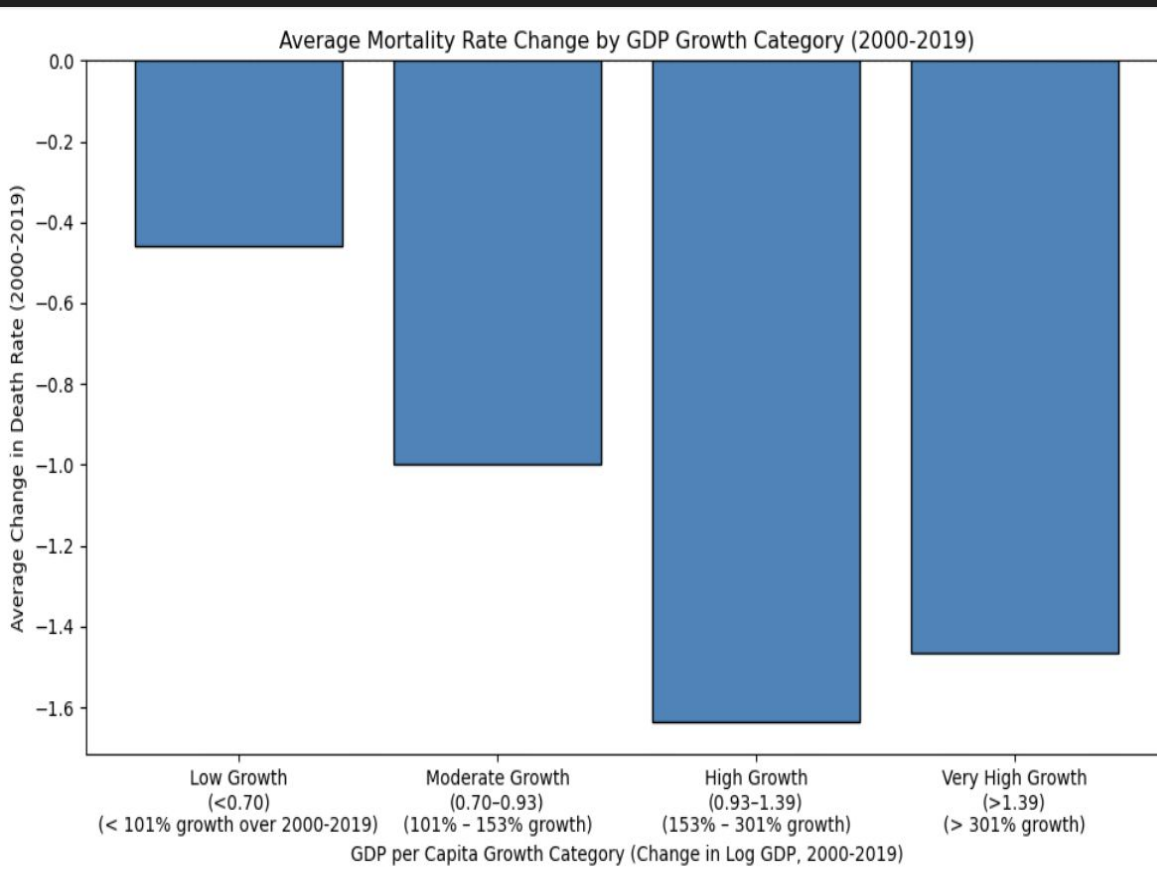
- Approximately symmetric - modest changes
- Mean = -1.14%, SD = 3.26%
- 14 outliers - mostly African countries & Monaco
- Africa - large decreases in mortality rates, rapid healthcare advancements
- Monaco - large increase in mortality rates, increase in elderly population
- Retain outliers - genuine variation across countries, not measurement error

Graph 2: Boxplot with Mortality Rates (2000 vs. 2019)



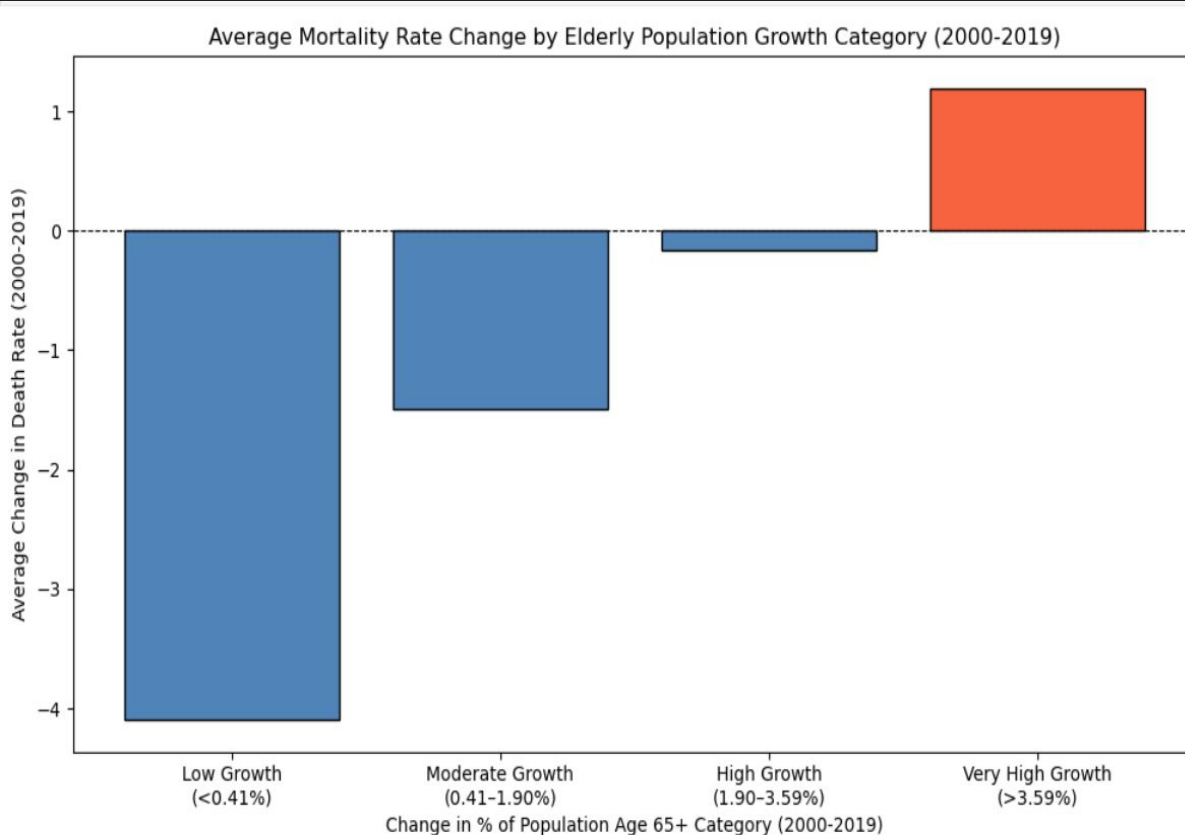
- 2000: Median = 8.5%, kurtosis = -0.109
- 2019: Median = 7.41%, kurtosis = 11.70
- Mortality rates declined overall (median decrease)
- But variation across countries increased a lot (kurtosis increase)
- Extreme outliers = very high mortality rates
- Ex. Sierra Leone (SLE) - 20% mortality rate in 2000
- Ex. Central African Republic - 30.25% mortality rate in 2019

Graph 3: Bar Graph with Mortality Rate Changes By GDP Growth (2000–2019)



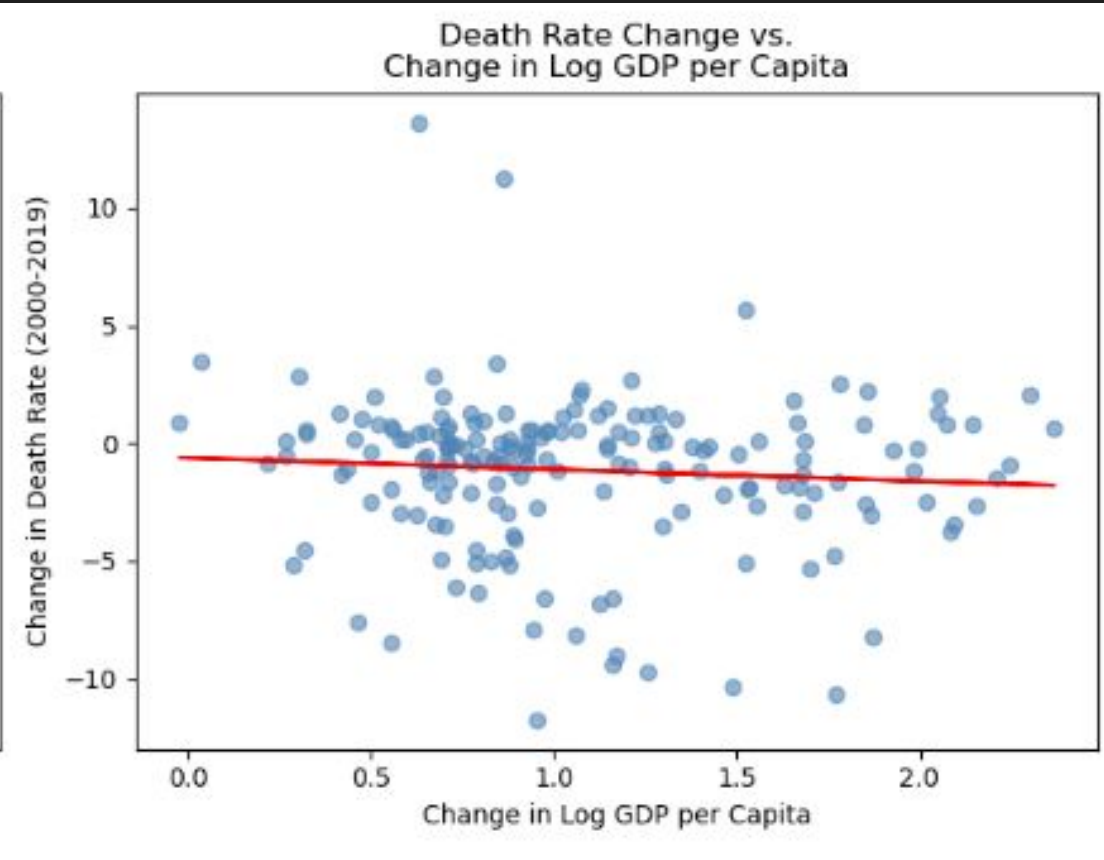
- Low → high growth = average change in mortality rates decreases
- Economy grow more = greater decrease in mortality rates
- Very high growth: decrease in mortality rates not as big
- Very high GDP growth & other factors (e.g. rise in elderly population) occur together
- Can't tell which factor truly led to less big decrease in mortality rates

Graph 4: Bar Graph with Mortality Rate Changes By Elderly Population (2000–2019)



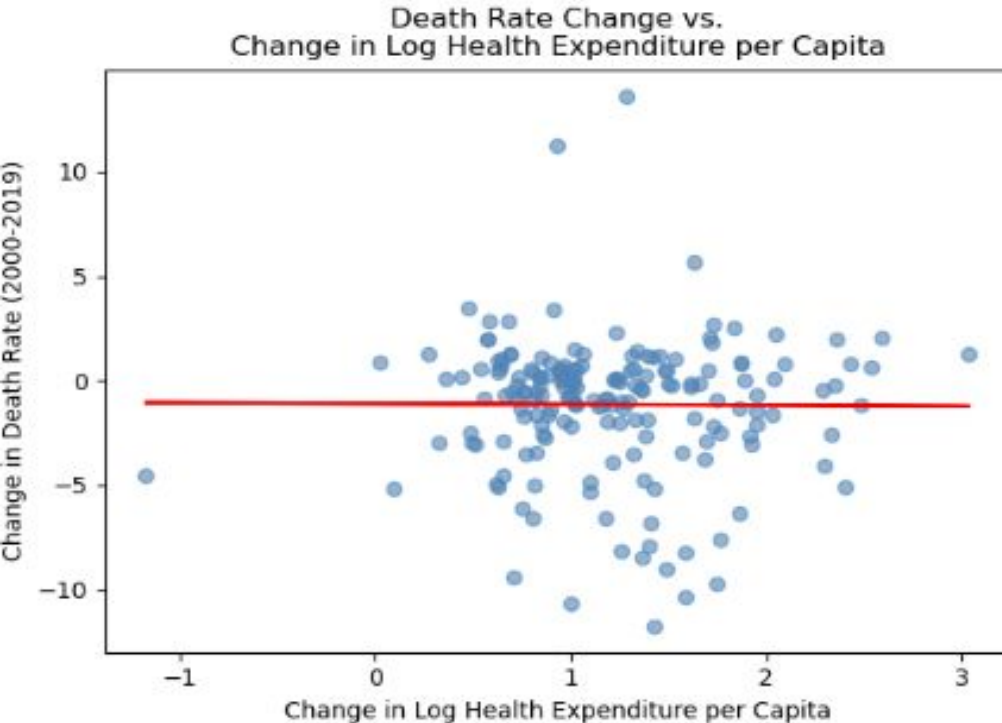
- Low → high growth = average change in mortality rates increases
- Elderly population grow more = greater increase in mortality rates
- Very high growth: positive change in mortality rates (very big increase)
- Very high elderly population growth & other factors (e.g. healthcare system quality) offset each other
- Impacts different for low v. very high growth countries

Graph 5A: Scatterplot with Changes in Death Rate vs. Changes in Log GDP Per Capita (2000–2019)



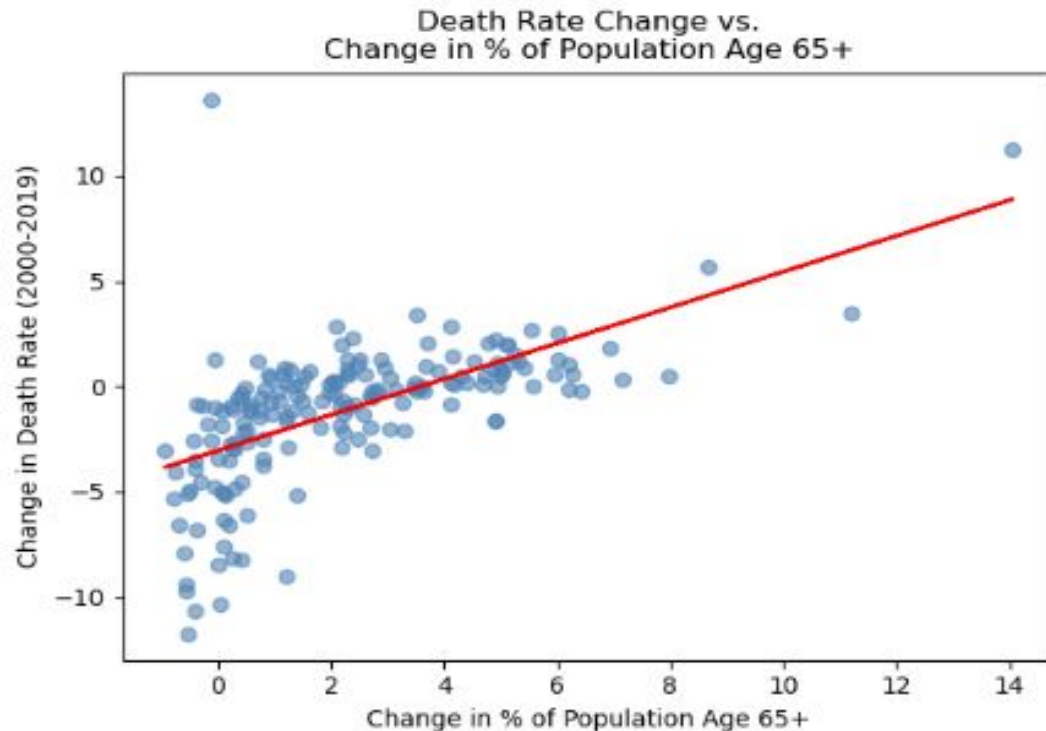
- Used logs - linear relationship, growth relative to actual size
- Negative relationship - GDP per capita grow = death rate decrease
- Connects to findings in Graph 3

Graph 5B: Scatterplot with Changes in Death Rate vs. Changes in Log Health Expenditure Per Capita (2000–2019)



- Used logs - linear relationship, growth relative to actual size
- No relationship - randomly scattered points, flat trendline
- Health expenditure per capita grow = death rate not impacted
- Lag effects - long time to see if health expenditure truly reduced mortality rates
- Diminishing returns - wealthy countries decrease mortality rates less than poorer ones by increasing health expenditure

Graph 5C: Scatterplot with Changes in Death Rate vs. Changes in % of Population Age 65+ (2000–2019)



- No logs - symmetric, less outliers (similar to graph 1 findings)
- Positive relationship - elderly population grow = death rate increases
- Supports graph 4
- Elderly → weaker immune system → more susceptible to disease → higher mortality rates

EDA Takeaways

- Strongest predictor = change in proportion of the elderly population
- GDP growth = weaker predictor
- Direction of relationships (e.g. positive/negative) makes business sense for GDP & elderly population
- Health expenditure change unrelated to mortality changes



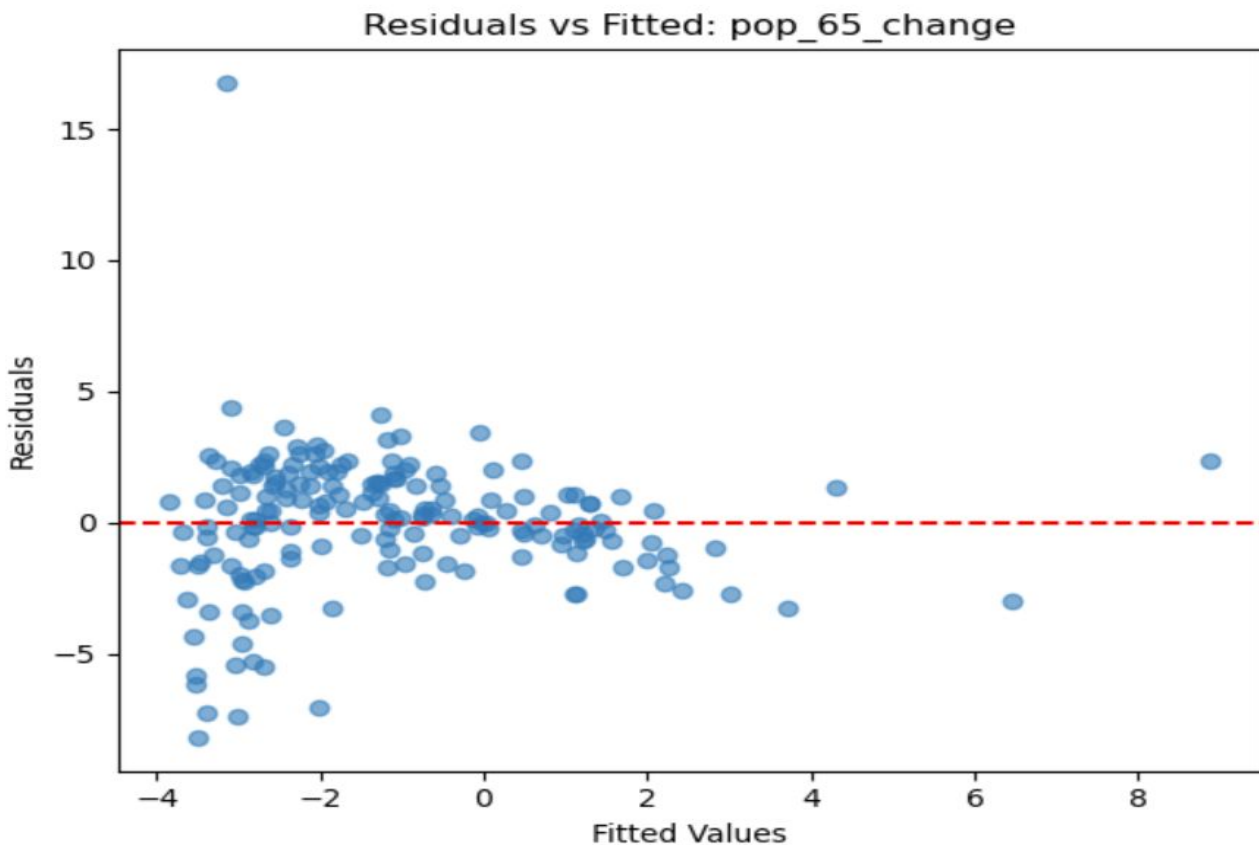
Modeling: Correlation Analysis

	death_change	log_gdp_change	log_health_change	\
death_change	1.000000	-0.077542	-0.006860	
log_gdp_change	-0.077542	1.000000	0.781329	
log_health_change	-0.006860	0.781329	1.000000	
pop_65_change	0.614353	0.008515	0.107532	

	pop_65_change
death_change	0.614353
log_gdp_change	0.008515
log_health_change	0.107532
pop_65_change	1.000000

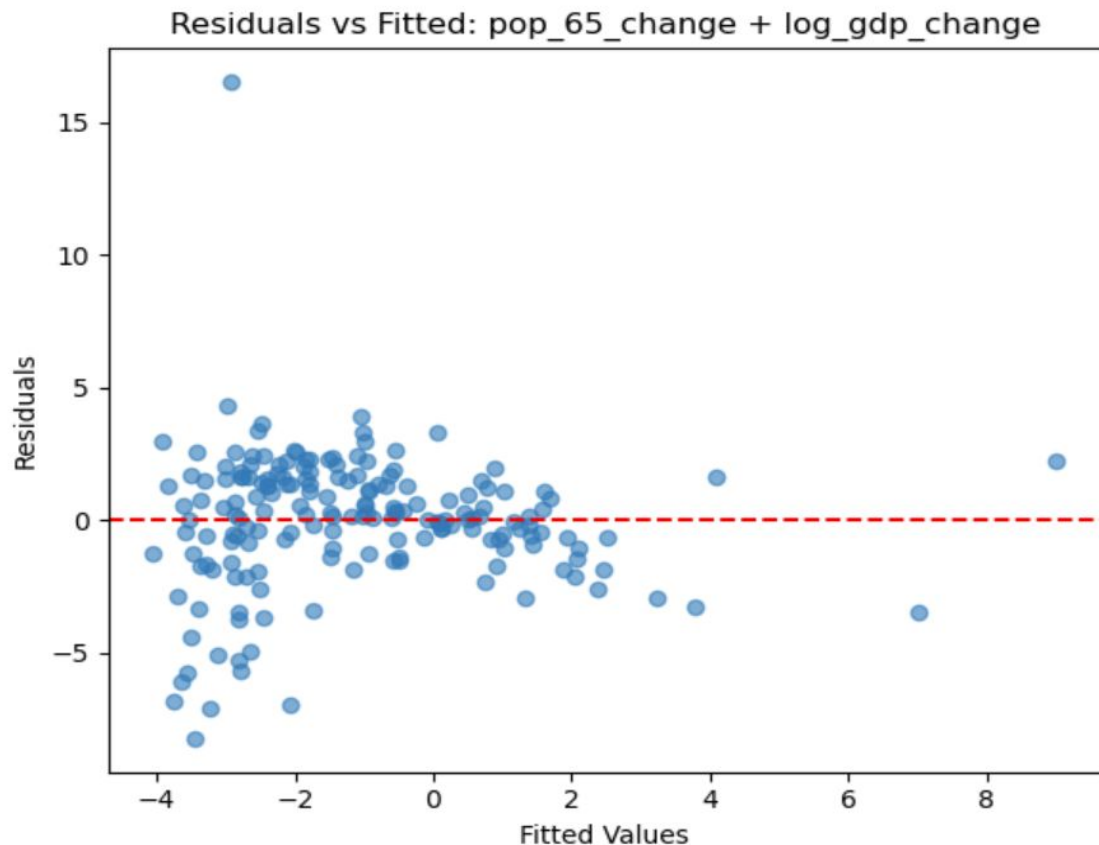
- Change in death rates & change in log GDP per capita = $-0.08 \rightarrow$ weak, negative
- Change in death rates & change in log health expenditure per capita = $-0.007 \rightarrow$ nonexistent, bad predictor
- Change in death rates & change in elderly population = 0.61 , strong, positive

Modeling: Assumptions & Methodology I



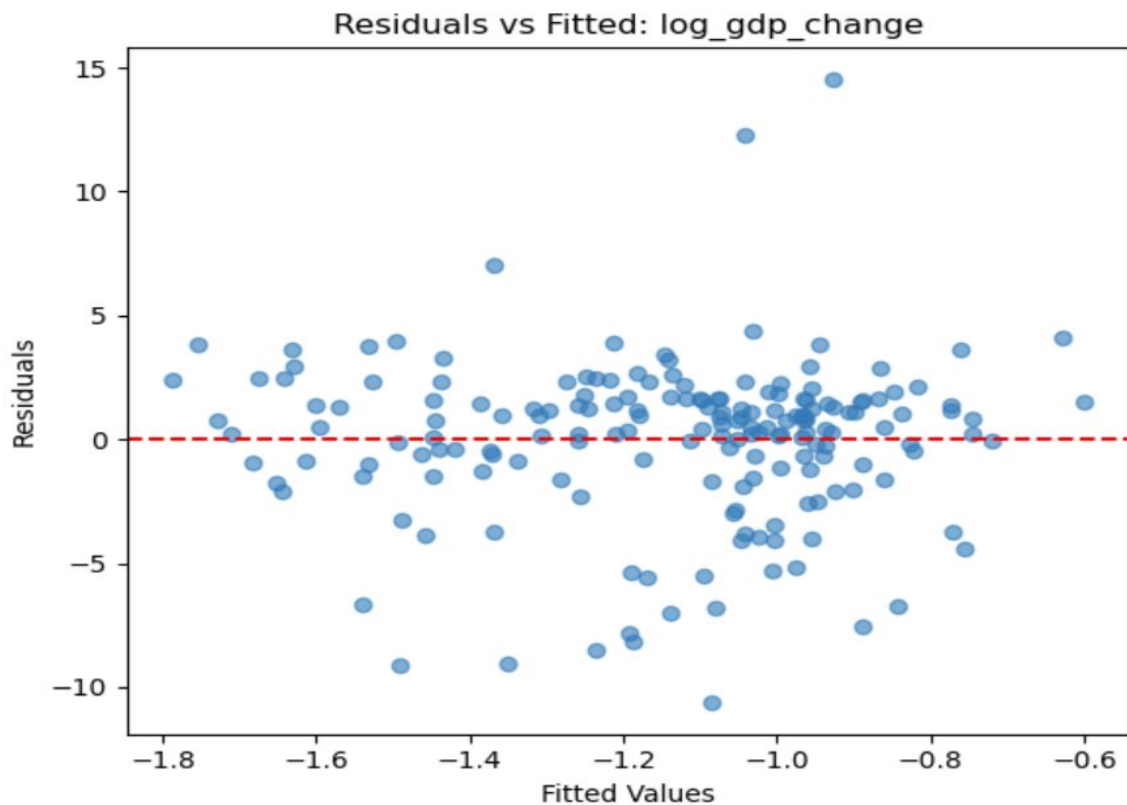
- OLS regressions: normally distributed residuals, constant variances for residuals (no heteroskedasticity), & linear relationships
- Residual plots - pop_65_change = non-random, tilted U-shape - non-linear relationships

Modeling: Assumptions & Methodology II



- OLS regressions: normally distributed residuals, constant variances for residuals (no heteroskedasticity), & linear relationships
- Residual plots - multivariate linear regression
(pop_65_change + log_gdp_change) = non-random, funnel shape - non-linear relationships

Modeling: Assumptions & Methodology III



- OLS regressions: normally distributed residuals, constant variances for residuals (no heteroskedasticity), & linear relationships
- Residual plots -
log_gdp_change = non-random, funnel shape- non-linear relationship

Residuals vs Fitted: pop_65_change + log_gdp_change

Modeling: Assumptions & Methodology IV

- Linear regression & single predictor – p-value & R-Sq. most important
- Polynomial regression – Adj. R-Sq. & p-value of highest order term
- P-values – 5% statistical significance
- Adj. R-Sq. – ≥ 0.05 increase for meaningful improvement in predictive power – more rigorous than R-Sq. = penalize model for complexity/added term



Modeling: Regression 1 Analysis

OLS Regression Results

```
=====
Dep. Variable:      death_change    R-squared:                0.006
Model:              OLS             Adj. R-squared:           0.000
Method:             Least Squares   F-statistic:              1.089
Date:               Mon, 27 Apr 2026 Prob (F-statistic):       0.298
Time:               20:44:53         Log-Likelihood:           -472.51
No. Observations:   182             AIC:                     949.0
Df Residuals:       180             BIC:                     955.4
Df Model:            1
Covariance Type:    nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	-0.6135	0.562	-1.092	0.276	-1.722	0.495
log_gdp_change	-0.4961	0.475	-1.043	0.298	-1.434	0.442

```
=====
Omnibus:            22.240          Durbin-Watson:           1.912
Prob(Omnibus):       0.000          Jarque-Bera (JB):         97.048
Skew:                -0.200         Prob(JB):                 8.44e-22
Kurtosis:            6.555          Cond. No.                 4.49
=====
```

- P-value = 0.298 – NOT statistically significant
- R-Sq. = 0.006 – 0.6% of variation in changes in death rates explained by $\log(\text{GDP per capita})$
- Conclusion: high p-value & low R-Sq. = `log_gdp_change` is a weak predictor

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Modeling: Regression 2 Analysis

OLS Regression Results

```

=====
Dep. Variable:      death_change    R-squared:                0.019
Model:              OLS             Adj. R-squared:           0.008
Method:             Least Squares   F-statistic:             1.757
Date:               Mon, 27 Apr 2026 Prob (F-statistic):       0.175
Time:               20:44:53        Log-Likelihood:          -471.29
No. Observations:   182            AIC:                     948.6
Df Residuals:       179            BIC:                     958.2
Df Model:           2
Covariance Type:    nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
const	0.8861	1.115	0.795	0.428	-1.314	3.087
log_gdp_change	-3.5512	2.021	-1.757	0.081	-7.540	0.437
log_gdp_change_sq	1.2596	0.810	1.555	0.122	-0.339	2.858

```

=====
Omnibus:            21.612    Durbin-Watson:           1.960
Prob(Omnibus):      0.000    Jarque-Bera (JB):         96.845
Skew:               -0.138    Prob(JB):                 9.34e-22
Kurtosis:           6.563    Cond. No.                  23.6
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

- Adj. R-Sq. = 0.008 – 0.8% of variation in changes in death rates explained by 1st & 2nd order terms for change in log(GDP per capita) after penalizing model
- NOT meaningfully higher than Adj. R-Sq. of 0 for Regression #1
- Conclusion: Adj. R-Sq. not increase much = discard polynomial model

Modeling: Regression 3 Analysis

OLS Regression Results

```
=====
Dep. Variable:      death_change  R-squared:              0.377
Model:              OLS          Adj. R-squared:            0.374
Method:             Least Squares  F-statistic:           109.1
Date:               Mon, 27 Apr 2026  Prob (F-statistic):      2.87e-20
Time:               20:44:53      Log-Likelihood:         -429.94
No. Observations:   182          AIC:                     863.9
Df Residuals:       180          BIC:                     870.3
Df Model:           1
Covariance Type:    nonrobust
=====
```

```
=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
const        -3.0461      0.264     -11.525      0.000     -3.568     -2.525
pop_65_change  0.8496      0.081      10.446      0.000      0.689      1.010
=====
```

```
=====
Omnibus:                57.681  Durbin-Watson:                2.084
Prob(Omnibus):           0.000  Jarque-Bera (JB):          744.689
Skew:                    0.716  Prob(JB):                  1.96e-162
Kurtosis:                12.806  Cond. No.                   4.70
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

- P-value = 0 – statistically significant
- R-Sq. = 0.377 – 37.7% of variation in changes in death rates explained by changes in proportion of elderly population
- Conclusion: low p-value & higher R-Sq. = pop_65_change is a strong predictor

Modeling: Regression 4 Analysis

OLS Regression Results

```
=====
Dep. Variable:      death_change    R-squared:                0.384
Model:              OLS             Adj. R-squared:           0.377
Method:             Least Squares   F-statistic:              55.87
Date:               Mon, 27 Apr 2026 Prob (F-statistic):       1.40e-19
Time:               20:44:53        Log-Likelihood:           -428.93
No. Observations:   182            AIC:                     863.9
Df Residuals:       179            BIC:                     873.5
Df Model:           2
Covariance Type:    nonrobust
=====
```

```
=====
              coef    std err          t      P>|t|      [0.025    0.975]
-----
const          -3.2240      0.292    -11.041     0.000     -3.800     -2.648
pop_65_change    1.0546      0.166     6.351     0.000      0.727      1.382
pop_65_change_sq -0.0267      0.019    -1.415     0.159     -0.064      0.011
=====
```

```
=====
Omnibus:                 65.881    Durbin-Watson:                2.073
Prob(Omnibus):            0.000    Jarque-Bera (JB):              841.817
Skew:                     0.912    Prob(JB):                      1.59e-183
Kurtosis:                 13.377    Cond. No.                      38.9
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

- Adj. R-Sq. = 0.377 – 37.7% of variation in changes in death rates explained by 1st & 2nd order terms for change in proportion of elderly population after penalizing model
- NOT meaningfully higher than adj. R-Sq. of 0.374 for Regression #3
- Conclusion: Adj. R-Sq. not increase much = discard polynomial model

Modeling: Regression 5 Analysis

OLS Regression Results

```
=====
Dep. Variable:      death_change    R-squared:                0.302
Model:              OLS             Adj. R-squared:           0.298
Method:             Least Squares   F-statistic:              77.78
Date:               Mon, 27 Apr 2026 Prob (F-statistic):       9.80e-16
Time:               20:44:53         Log-Likelihood:           -440.38
No. Observations:   182             AIC:                      884.8
Df Residuals:       180             BIC:                      891.2
Df Model:           1
Covariance Type:    nonrobust
=====
```

```
=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
const         -0.7907      0.207      -3.826      0.000      -1.198      -0.383
log_pop_65_change  0.6407      0.073       8.819      0.000       0.497       0.784
=====
```

```
=====
Omnibus:                 95.746    Durbin-Watson:                1.984
Prob(Omnibus):            0.000    Jarque-Bera (JB):            1377.728
Skew:                     1.553    Prob(JB):                     6.76e-300
Kurtosis:                 16.116    Cond. No.                     2.91
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

- R-Sq. = 0.302 – 30.2% of variation in changes in death rates explained by log of the change in proportion of elderly population
- Much smaller than R-Sq. of 0.374 for regression #3
- Conclusion: Logarithmic predictor explains much less variation = discard model

Modeling: Regression 5 Analysis

- Why Logarithms failed?
- Input values $\geq 0 \rightarrow$ countries with 0 or negative growth forced to have 0.001% growth
- Country with 3% decline = Country with 0% growth - inaccurate!
- Does not realize growth can be positive, negative, or 0



Modeling: Regression 6 Analysis

OLS Regression Results

```
=====
Dep. Variable:      death_change    R-squared:                0.384
Model:              OLS             Adj. R-squared:           0.377
Method:             Least Squares   F-statistic:              55.86
Date:               Mon, 27 Apr 2026 Prob (F-statistic):       1.41e-19
Time:               20:44:53         Log-Likelihood:           -428.93
No. Observations:   182             AIC:                      863.9
Df Residuals:       179             BIC:                      873.5
Df Model:           2
Covariance Type:    nonrobust
=====
```

```
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const          -2.4833      0.478      -5.195      0.000      -3.427      -1.540
pop_65_change    0.8505      0.081     10.487      0.000       0.690       1.011
log_gdp_change  -0.5296      0.375     -1.411      0.160      -1.270       0.211
=====
```

```
=====
Omnibus:                55.459    Durbin-Watson:                2.089
Prob(Omnibus):           0.000    Jarque-Bera (JB):             690.418
Skew:                    0.676    Prob(JB):                     1.20e-150
Kurtosis:                12.445    Cond. No.                     10.5
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

- Adj. R-Sq. = 0.377 – 37.7% of variation in changes in death rates explained by changes in proportion of elderly population & change in log GDP per capita after penalizing
- NOT much bigger than Adj. R-Sq. of 0.374 for regression #3
- Conclusion: Adj. R-Sq. NOT meaningfully increase = discard log_gdp_change in model, revert to regression #3

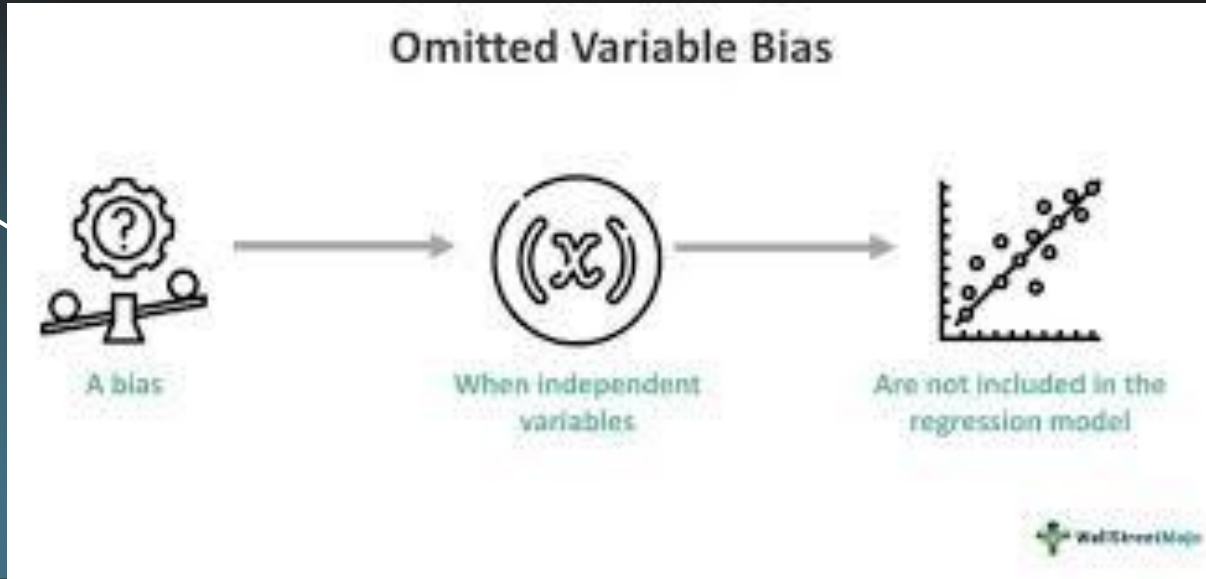
Regression Analysis Summary & Main Findings

Regression #	Predictors	Coefficient	P-value	R ²	Adj. R ²
1	log_gdp_change (simple linear)	-0.4961	0.298	0.006	0
2	log_gdp_change_squared (polynomial)	_____	• 0.081 (log_gdp_change) • 0.122 (log_gdp_change_squared)	0.019	0.008
3	pop_65_change (simple linear)	0.8496	0	0.377	0.374
4	pop_65_change_squared	_____	• 0 (pop_65_change) • 0.159 (pop_65_change_squared)	0.384	0.377
5	log_pop_65_change (logarithmic)	0.6407	0	0.302	0.298
6	pop_65_change + log_gdp_change (multivariate linear)	• 0.8505 (pop_65_change) • -0.5296 (log_gdp_change)	• 0 (pop_65_change) • 0.16 (log_gdp_change)	0.384	0.377

- Best Model: Regression #3
- Coefficient = 0.8496 – As the change in the proportion of the elderly population increases by 1 percentage point, then the change in mortality rates increases by 0.8496 percentage points.
- NOT significant – only explains 37.4% of variation for changes in death rates

Limitations

- Omitted variable bias – country can have BOTH large elderly population & high quality healthcare (not in model) – both affect mortality rate changes
- Higher quality healthcare reduces mortality rate, increased elderly population increases mortality rate
- Cannot tell if changes in elderly population caused changes in mortality rates – confounding



Reflection

What worked well

- Correlation matrix → discard health expenditure (insignificant) – saves time
- Testing multiple models (linear, polynomial, logarithmic) – confirmed linear regression best for pop_65_change

Improvements

- Incorporate different predictors to reduce omitted variable bias
- Many variables driving mortality rate changes
- Ex. Healthcare system quality



AI Acknowledgement

I used AI to help write the code for SQL queries, graphs, and regressions, and it also helped with proofreading my analyses and brainstorming ideas.





Thank you!